

- 4 (a) ability to do work as a result of the position/shape, etc. of an object
- B1
B1 [2]

(b) (i) 1 $\Delta E_{\text{gpe}} = GMm / r$ C1
 $= (6.67 \times 10^{-11} \times \{2 \times 1.66 \times 10^{-27}\}^2) / (3.8 \times 10^{-15})$ C1
 $= 1.93 \times 10^{-49} \text{ J}$ A1 [3]

2 $\Delta E_{\text{epe}} = Qq / 4\pi\epsilon_0 r$ C1
 $= (1.6 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times 3.8 \times 10^{-15})$ C1
 $= 6.06 \times 10^{-14} \text{ J}$ A1 [3]

(ii) idea that $2E_K = \Delta E_{\text{epe}} - \Delta E_{\text{gpe}}$ B1
 $E_K = 3.03 \times 10^{-14} \text{ J}$
 $= (3.03 \times 10^{-14}) / 1.6 \times 10^{-13}$ M1
 $= 0.19 \text{ MeV}$ A0 [2]

- (iii) fusion may occur / may break into sub-nuclear particles B1 [1]

- 5 (a) (i) V_{c} depends on angle between (plane of) probe and B field

B1